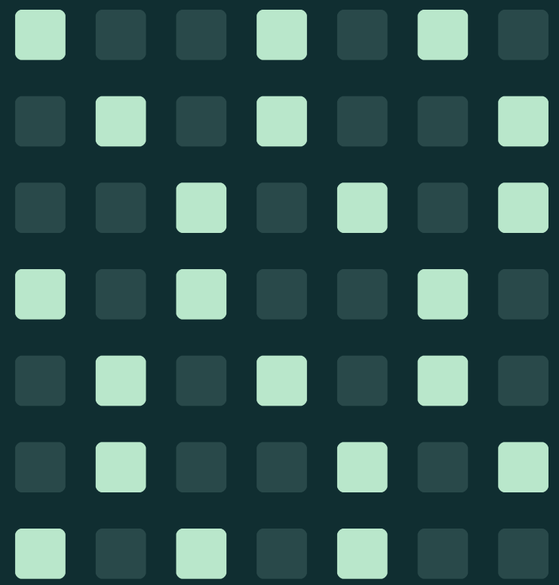
Watermark Lab

Statistical signals in AI writing

Insertion, detection and survival
under recursive paraphrasing



A pattern in token choices

Completed local study
23 October 2026

Research prototype and completed benchmark

The problem and the reason for this study

Can a deliberate signal survive when the wording changes?

01

Insertion

The generator favors a secret set of token choices.

02

Rewriting

The attacker changes wording without our processor.

03

Detection

The detector reconstructs the pattern with the same key.

Baseline rationale: no retraining, explicit detection and measurable parameter tradeoffs

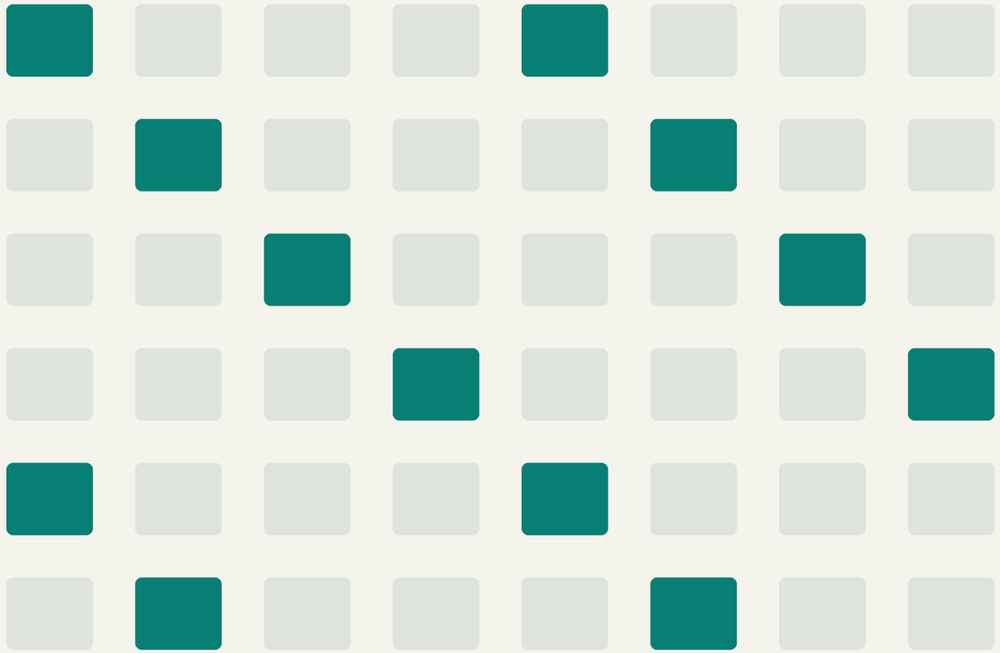
The watermark changes next-token sampling

Key + previous token

HMAC-SHA256 seeds a deterministic vocabulary partition through PCG64.

Approximately 25% favored

Add delta to green logits.
The model still samples among alternatives.



Eligible next-token vocabulary

Same key and tokenizer reconstruct the favored choices during detection

The score measures excess favored choices

$$z = (G - \text{gamma} \times T) / \sqrt{[T \times \text{gamma} \times (1 - \text{gamma})]}$$

G = favored choices T = checked evidence units

Marked original	116 / 195	**11.12**
First rewrite	48 / 194	**–0.08**

Cutoff = 4. Below cutoff means insufficient evidence for this pattern.

The app connects writing to the evidence

01 **Generate & compare**

Ordinary and marked passages

02 **Check a passage**

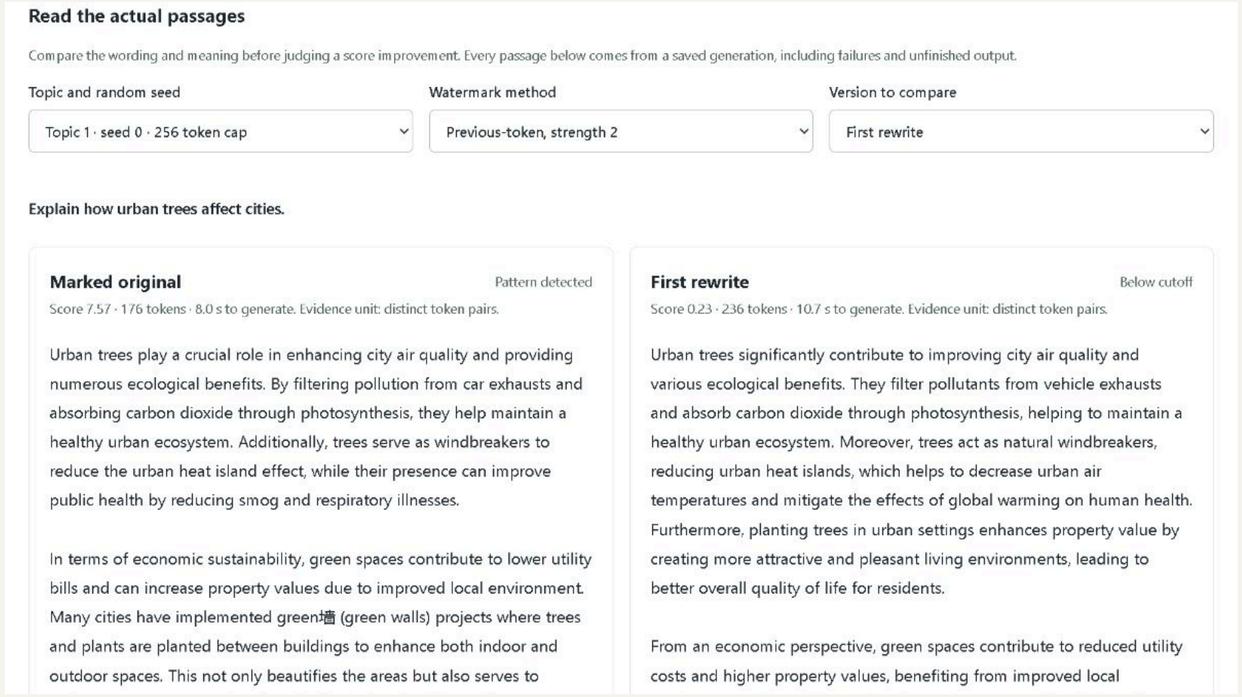
Rebuild the keyed pattern

03 **Try rewriting**

Three recursive paraphrases

04 **Research bench**

Rates, reviews and checkpoints



Actual saved paragraph-comparison layout

Single-passage lab: localhost:8765 Research bench: localhost:8766

The controlled local experiment

20 topics × 3 seeds × 3 methods

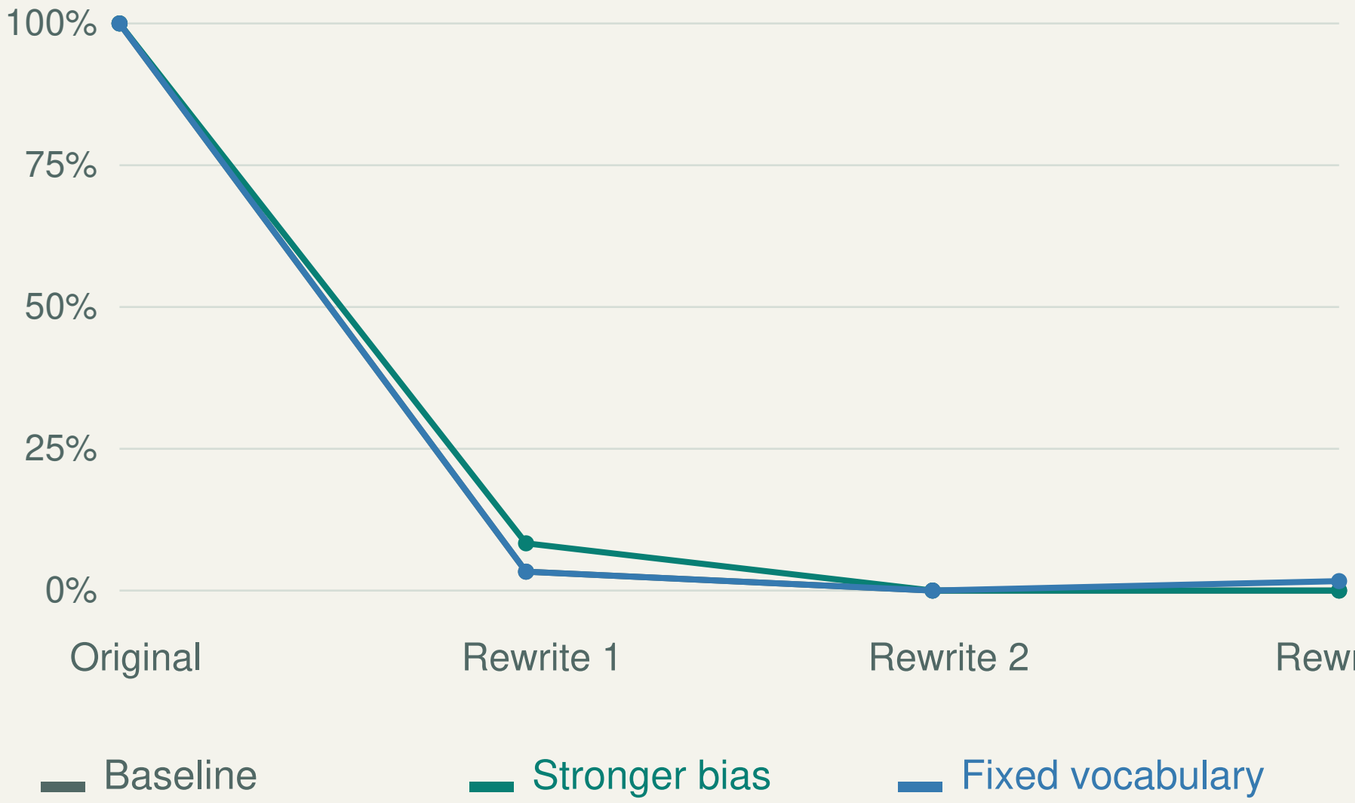
180 main cells, each with five scored conditions

Method	Partition	Bias
Baseline	Previous token	delta 2
Stronger bias	Previous token	delta 3
Fixed vocabulary	Constant partition	delta 3

Length sweep: 30 more cells at caps 128 and 512

210 completed cells and 1,050 rows. Ordinary control text is shared across methods.

Rewriting removes most detectable evidence



Method	After 1	After 3
Baseline	2/60	0/60
Stronger bias	5/60	0/60
Fixed vocabulary	2/60	1/60

Originals: 60/60 per method. Controls: 0/60 triggers. Human false-positive rate remains unmeasured.

Research foundations and alternative methods

Green-list insertion

Kirchenbauer et al. / ICML 2023

Soft promotion and independent detection

Retention after editing

Kirchenbauer et al. / ICLR 2024

Residual fragments and observed text length

Recursive paraphrase attacks

Sadasivan et al. / TMLR

Stress tests and spoofing concerns

Fixed vocabulary grouping

Zhao et al. / ICLR 2024

Unigram-Watermark and conditional guarantees

Semantic context

Liu et al. / ICLR 2024

SIR embedding and trained transformation

Distortion-free sampling

Kuditipudi et al. / TMLR 2024

Keyed sequence alignment for detection

Production evaluation

Dathathri et al. / Nature 2024

SynthID-Text quality and efficiency

Joint quality and detection

Tu et al. / ACL 2024

WaterBench comparison protocol

Full primary-source citations in speaker notes. Published findings and our measurements have separate scopes.

The quality and validation gaps

Originals reaching the token cap

Method	Cap flags
Baseline	9/60
Stronger bias	25/60
Fixed vocabulary	35/60

Review still required

- Meaning and claims preservation
- Human false-positive calibration
- Independent paraphrasers and keys
- Faithful semantic comparison

No validated robust winner yet

Cap flags do not equal quality ratings. Current human-review count: 0.

The next milestone is a fair robustness gain

More faithful rewrites detected, at a controlled error rate

01

Quality and calibration

Review saved passages and collect human negatives.

02

Faithful alternative

Integrate the semantic encoder and reference method.

03

Independent confirmation

Test new attacks, keys and models on fresh cases.

Completed: insertion, inspection and local benchmark. Next: quality-aware comparison.